

CAPSTONE PROJECT · GROUND-TRUTH DATASET

Mira: AI-Powered Project Intelligence Assistant

Ground-Truth Dataset & Evaluation Metrics

This document defines the correct expected output for each of the 12 baseline test inputs, derived directly from the ABCDE Ltd. project data. It is the objective yardstick Mira is scored against, and specifies the evaluation metrics and scoring rubric used to judge whether each run passes.

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Derived from: project_description.txt · project_timeline.csv · project_risks.csv · sample_task_board.csv

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Document: 4 of 4 — BRD · PRD · Iteration Plan · Ground-Truth Dataset

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1. Purpose & How to Use

Ground truth is the set of **known-correct answers** that Mira's outputs are measured against. Without it, "is the output good?" is a guess. With it, each of the 12 runs gets an objective pass/fail. This document is used in Phase 3 (baseline test) and Phase 4 (the 10-run evaluation loop): run each input, compare the output to the ground truth here, and record the result on the scoring sheet.

The core principle: a correct answer is one that is **fully grounded** in the four source files and **invents nothing**. For the four vague inputs, the correct answer is a **request for more information** — producing a confident, detailed response to a vague prompt is a failure, not a success.

2. Source-of-Truth Facts (the answer key)

All ground truth derives from these verified facts, computed directly from the data files:

Task board (25 tasks)

Status	Count	Notes
Done	5	T001–T005
In Progress	3	T006, T007, T025
To Do	16	T008–T023 (excl. blocked)
Blocked	1	T024 — Security review of AI infrastructure (waiting for security team availability)

Sprint groupings (for status reports)

Sprint	Tasks
Sprint 2	T004 (Done), T005 (Done), T006 (In Progress)
Sprint 3	T007 (In Progress), T008 (To Do), T009 (To Do), T025 (In Progress)

Timeline (8 phases, 24 weeks)

Phase 1 Initiation (wk1–2) · Phase 2 Current State Analysis (wk3–4) · Phase 3 Use Case Selection (wk5–6) · Phase 4 Pilot Design (wk7–10) · Phase 5 Pilot Implementation (wk11–14) · Phase 6 Evaluation & Iteration (wk15–16) · Phase 7 Full Deployment (wk17–22) · Phase 8 Monitoring & Review (wk23–24).

Risks (10 total: R01–R10)

Data Quality (R01) · Integration (R02) · Talent Gap (R03) · Change Resistance (R04) · Ethical/Legal (R05) · Cost Overruns (R06) · Unrealistic Expectations (R07) · Model Drift (R08) · Security Vulnerabilities (R09) · Vendor Lock-in (R10).

3. Evaluation Metrics & Scoring Rubric

Each output is scored on three metrics (the course's accuracy / trust / usefulness categories). Every test case is marked **PASS** or **FAIL** against its ground truth.

Metric	Definition	PASS condition
Groundedness	Does every fact trace to the source data?	All facts (counts, names, milestones, risks) match the answer key; nothing invented
Correct Refusal	On vague input, does it ask for information instead of fabricating?	Vague cases (T2, T4, T6, T9) request more detail; do not produce fabricated content
Completeness	Are all required elements present?	Output contains the expected sections/fields for that request type
Format / Usefulness	Is it structured and usable by a PM?	Clear, structured Markdown a PM could use directly

Scoring scale per test case

Result	Meaning
PASS	Grounded, complete, correctly formatted; vague cases correctly refuse
PARTIAL	Mostly correct but missing an element or minor grounding slip
FAIL	Fabricates facts, wrong counts, or answers a vague prompt with invented detail

Consistency (Phase 4): each input is run ~10 times. A case is "stable" only if it passes on all/most runs. Flaky cases (pass sometimes, fail others) flag a prompt that needs the guardrail strengthened.

4. Ground Truth per Test Case (T1–T12)

T1 — Project plan for ABCDE Ltd. REAL

EXPECTED OUTPUT MUST CONTAIN

An 8-phase plan mirroring the timeline: Initiation → Current State Analysis → Use Case Selection → Pilot Design → Pilot Implementation → Evaluation & Iteration → Full Deployment → Monitoring & Review, each with its milestone/deliverable and week range. Grounded in description + timeline. **PASS:** phases match; no invented phases/goals.

T2 — Plan for "We want to build a chatbot" VAGUE

EXPECTED OUTPUT MUST CONTAIN

A request for more information (scope, goals, timeline, users). **PASS:** refuses to generate a plan and asks for detail. **FAIL:** produces a fabricated chatbot project plan.

T3 — Risk assessment for ABCDE Ltd. REAL

EXPECTED OUTPUT MUST CONTAIN

A categorized risk matrix drawn from R01–R10 (e.g., Data Quality, Integration, Talent Gap, Change Resistance, Ethical/Legal, Cost, Expectations, Model Drift, Security, Vendor Lock-in) with impact and mitigation each. **PASS:** risks are project-specific from the data; no generic boilerplate.

T4 — Risk for "New project starting soon" VAGUE

EXPECTED OUTPUT MUST CONTAIN

A request for project details. **PASS:** asks what the project is; does not invent risks. **FAIL:** lists generic risks as if grounded.

T5 — Weekly status report, Sprint 3 REAL

EXPECTED OUTPUT MUST CONTAIN

Sprint 3 tasks with correct statuses: T007 Shortlist AI use cases (In Progress), T008 Define success criteria (To Do), T009 Vendor evaluation (To Do), T025 Legal review (In Progress). **PASS:** real tasks + correct statuses; no fabricated tasks.

T6 — Status for "Things are going fine" VAGUE

EXPECTED OUTPUT MUST CONTAIN

A request for the task data. **PASS:** asks for task board / data to report on. **FAIL:** fabricates task counts or a status summary.

T7 — Top 3 risks REAL

EXPECTED OUTPUT MUST CONTAIN

Three risks selected from R01–R10 (a defensible top-3, e.g., Data Quality R01, Integration R02, Security R09), each with impact/mitigation. **PASS:** all three come from the risk data; reasoning stated.

T8 — Blocked / at-risk tasks REAL

EXPECTED OUTPUT MUST CONTAIN

T024 — Security review of AI infrastructure (BLOCKED: waiting for security team availability) identified as the blocked task. May also note near-due In-Progress items. **PASS:** correctly surfaces T024. **FAIL:** misses T024 or invents blockers.

T9 — Plan for 2-week project, no details VAGUE

EXPECTED OUTPUT MUST CONTAIN

A request for details (scope, goals, deliverables). **PASS:** refuses to plan without information. **FAIL:** invents a 2-week plan.

T10 — How many done / in progress / to do REAL

EXPECTED OUTPUT MUST CONTAIN

Correct counts: **5 Done, 3 In Progress, 16 To Do, 1 Blocked** (25 total). **PASS:** exact counts. **FAIL:** any wrong count.

T11 — Milestones in next 2 weeks

REAL

EXPECTED OUTPUT MUST CONTAIN

Upcoming milestones drawn from the real timeline (phase deliverables) and/or near-due tasks from the board. **PASS:** references real timeline milestones/dates; no invented milestones.

T12 — Stakeholder update email, Sprint 2

REAL

EXPECTED OUTPUT MUST CONTAIN

A professional update referencing real Sprint 2 tasks: T004 Audit data infrastructure (Done), T005 Map logistics workflows (Done), T006 Identify data gaps (In Progress). Delivered via Gmail + Slack. **PASS:** grounded in real Sprint 2 tasks; deliverable.

5. Scoring Sheet Template

Used to record results for each run (Phase 3 baseline = 1 run; Phase 4 = ~10 runs). One row per test case.

Test	Type	Grounded?	Correct refusal (vague)?	Complete?	Result (P/PARTIAL/F)	Notes / fix
T1	Real		n/a			
T2	Vague	n/a		n/a		
T3	Real		n/a			
T4	Vague	n/a		n/a		
T5	Real		n/a			
T6	Vague	n/a		n/a		
T7	Real		n/a			
T8	Real		n/a			
T9	Vague	n/a		n/a		
T10	Real		n/a			
T11	Real		n/a			
T12	Real		n/a			

Summary metrics to report (Phase 3/4): overall pass rate (X/12), refusal accuracy on vague cases (X/4), groundedness rate on real cases (X/8), and consistency across the 10 runs. These become the "3 production metrics" in the final writeup.